A1 Week 2 Plan

General Carbon/Contracting Theory & Numerical Methods — the road to a *Management Science* submission

Where we are. Week 1 delivered the spine: the capacity-substitution channel $K_R^* > K_R^\circ$, the exact net-carbon inequality (C), the distribution-free cutoffs, the closed-form thresholds $(\bar{C}_M, C_R^\dagger, C_R^\circ, \bar{C}_M^b)$, a 15/15 machine-precision audit, and a 3-referee honesty pass — cross-validated by all four roles. An adversarial *Management Science* review puts us at ≈70–75% of the rigor bar. The gap is one keystone (a general-distribution, finite-optimum carbon theorem) plus robustness, a clean numerical narrative, and positioning. Week 2 closes it.

Scope (hard wall). A1 only: general theory + numerical methods. *Not* A2 (empirical calibration / real-data magnitudes), *not* B1 (reliability, welfare, social-vs-private pricing), *not* B2 (flexibility continuum, AI, storage, multi-grid, dynamics). Every task below is theory or numerical-method work on the A1 model.

Parallelization. The four roles are designed to run simultaneously and share only notation + the Week-1 closed forms. Role 1 (lead) carries the keystone; Roles 2–4 consume Role 1's conditions *as they land* but can start immediately on their independent components.

Role 1 — Lead General Theory (keystone; carries the heaviest load)

Mandate. Turn the headline from "conditional/conjectural for general F" into a proved, finite-optimum, general-distribution theorem. The paper's title depends on this role.

- **Existence & uniqueness of the interior optimum.** Prove coercivity of the profit Π and give a primitive condition on F (e.g. log-concave density, or monotone hazard) under which any interior critical point is the unique global maximizer — discharging the "interior optimum is assumed" gap and supplying Condition 2 ($\mathcal{H} \prec 0$) from primitives rather than eigenvalue scans.
- **Promote the shift-ratio condition to a primitive.** Replace the endogenous-window condition "$g(x) = F_{K_R}/f$ non-increasing on $(K_R^* - K_M^*, K_R^*)$" with a support-wide monotone-g (or hazard-rate) condition on F that *implies* it, and prove the implication. State it as a clean primitive hypothesis.
- **Prove general-F non-emptiness of $\{\Delta \geq 0\}$ (the keystone).** Under the primitive shift-ratio condition plus a tail/skew condition, prove carbon *falls somewhere* in C_R for any such F — upgrading Theorem 3(iii) from conjecture to theorem, or sharply characterizing the F-class for which it holds.
- **Finite-interior-optimum carbon win.** Exhibit a primitive condition under which carbon strictly falls at a *finite, interior* K_R^* (not the $K_R^*{=}\infty$ knife-edge corner), so the result is not an unbounded-capacity artifact.
- **State the main theorem; re-scope the abstract/title** to match exactly what (1)–(4) deliver, and write the economic-intuition narrative (why $R > c$ + negative demand–supply association ⇒ carbon falls) at journal quality.

Role 2 — Distributional Robustness Theory

Mandate. Close the cross-distribution robustness layer that Week 1 left at "fixed capacities."

- **Sign the capacity covariation** dK^*/dF along the shape class $\mathcal{U}$, converting the least-favorable result from a fixed-capacity statement into a genuine cross-F *least-favorable-distribution* theorem (or prove the obstruction precisely).
- **Robust certificate.** Turn the magnitude bounds into a guarantee $\Delta \geq 0$ valid for *every* $F \in \mathcal{U}$ with $C_R \leq C_R^\dagger$, making the closed-form condition a true distributionally-robust certificate.
- **Repair the** $\mathbb{E}[\textbf{fossil}^*] \to 0$ **limit** with a dominated-convergence / finite-K_R envelope argument, removing the a.s.-corner hand-wave in the threshold proof.
- **Position the robustness machinery** against the least-favorable-distribution / distributionally-robust-optimization literature (feeds Role 4's bibliography).

Role 3 — Numerical Methods & Reproducibility

Mandate. One coherent, reproducible numerical backbone that demonstrates the theory at a finite interior optimum.

- **One carbon-sign narrative.** Replace the scattered "2/32, 10/40" with a principled sweep over Role 1's primitive condition on a curated distribution family, showing carbon falls at a *finite interior* optimum — one consistent story across Roles 1/3/4.
- **Hessian certificate.** A reproducible script confirming $\mathcal{H} \prec 0$ on the interior path for Role 1's distribution class (backstops the existence/regularity proof).
- **Build the reversed-channel counterexample.** Construct a realized F with g increasing $\Rightarrow$ $K_R^* < K_R^\circ$ — the sharpness example Week 1 could not produce (algebraic sign-flip is not a realized example).
- **Solver hardening + one-command harness.** Replace finite-difference cofactors with analytic ones (kills the noisy knife-edge points), document tolerances, and ship a single `reproduce` command covering the 15 exact-formula checks plus the carbon sweep.

Role 4 — Consistency, Regimes & Positioning

Mandate. Make the four pieces one manuscript a referee reads as a single, consistent paper.

- **Unify the cross-role statements.** One Assumption 1 ($R > c + C_M$) everywhere, one carbon decomposition (Role 1's two-force split), and strip any out-of-scope miner/oligopoly material — a single notation table and regime map for the whole manuscript.
- **Complete the regime map** with the finite-optimum statement from Role 1, so small-FCL / interior / full-coverage and the boundedness axis (C_R) compose into one consistent theorem.
- **Write the literature section** (theory side: capacity investment & peak-load pricing, monotone comparative statics, interruptible/ demand-response load, distributionally-robust / least-favorable methods) and add the bibliography — currently zero citations.
- **Assemble & edit the manuscript:** move all "verified to 10^{-k}" out of proof bodies into the numerical section, self-contain the draft corrections, and align every abstract claim with the theorem it cites.

Dependencies & sequencing

All four start immediately. Role 1 delivers two primitives early — the *primitive shift-ratio condition* (unblocks Role 3's sweep and Role 2's class $\mathcal{U}$) and the *existence condition* (unblocks Role 2's capacity

covariation and Role 3's Hessian certificate). Roles 2–4 hand Role 4 their final statements for assembly in the last two days. Acceptance gate: a *Management Science* referee cannot reject on theory/rigor once Role 1 (keystone) and Role 4 (consistency + positioning) land; Roles 2–3 turn an acceptance into a strong one.